MAT 3007 – Optimization

Intro to Nonlinear Optimization

Lecture 16

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Recap

Start: A New Perspective on Solving LPs

- We now consider the simplex method from a new perspective.

An optimality conditions for LPs (in standard form):

1. Primal Feasibility: $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, $\boldsymbol{x} \geq \boldsymbol{0}$.
2. Dual Feasibility: $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$.
3. Complementarity: $x_i \cdot s_i = x_i \cdot (c_i - \boldsymbol{A}_i^\top \boldsymbol{y}) = 0$ for each i.

These conditions are necessary and sufficient for $\boldsymbol{x}$ and $\boldsymbol{y}$ being the optimal solutions of the primal and dual problems.

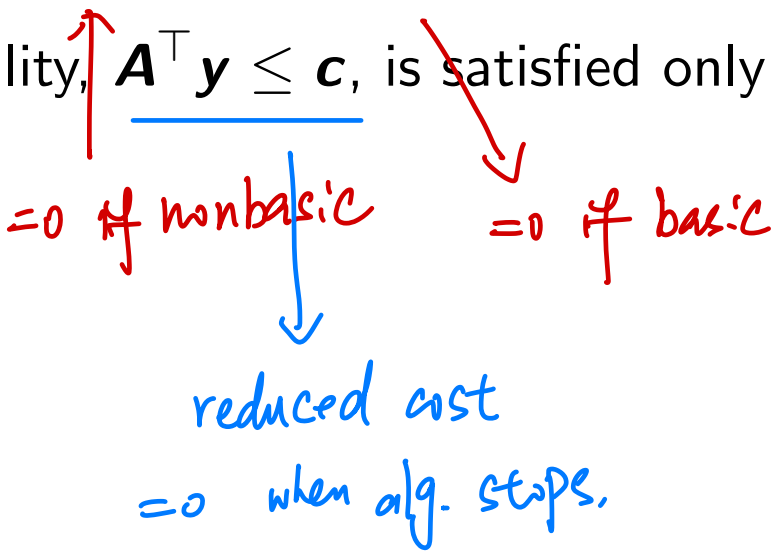
Review of the Simplex Method

In the simplex method, consider $\boldsymbol{y} = (\boldsymbol{A}_B^{-1})^\top \boldsymbol{c}_B$, for every iteration:

- We search among BFS, so it maintains primal feasibility.
- The complementarity condition is always satisfied:

$$x_i \cdot (c_i - \boldsymbol{A}_i^\top \boldsymbol{y}) = 0 \quad \forall\ i.$$

- The dual feasibility, $\boldsymbol{A}^\top \boldsymbol{y} \le \boldsymbol{c}$, is satisfied only when it stops.



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Proposition: Optimality Properties of the Simplex Method

During each iteration, the simplex method maintains primal feasibility and the complementarity conditions. It seeks a solution that is dual feasible.

Interior Point Method

The interior point method maintains primal feasibility and dual feasibility during its iterations and seeks for a pair of solutions that satisfy the complementarity conditions.

$$\begin{aligned} \boldsymbol{A}\boldsymbol{x} &= \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0} \\ \boldsymbol{A}^\top \boldsymbol{y} + \boldsymbol{s} &= \boldsymbol{c}, \quad \boldsymbol{s} \geq \boldsymbol{0} \\ x_i \cdot s_i &= \mu, \quad \forall\ i. \end{aligned}$$

s is dual slack

We call $\mu > 0$ the complementarity gap.

want to find a point s.t $\mu = 0$

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We call $\mu > 0$ the complementarity gap.

Idea: If we have found a solution for a certain μ, then we want to find a solution for a smaller μ. We keep decreasing μ.

$$\min \; c^T x$$
$$\text{s.t} \quad Ax = b$$
$$x \geq 0$$

$$\min \; c^T x - \mu \sum_{i=1}^{n} \log x_i$$

barrier function.

$$\text{s.t} \quad Ax = b$$

enforce $x_i > 0$

Why "Interior Point" Method?

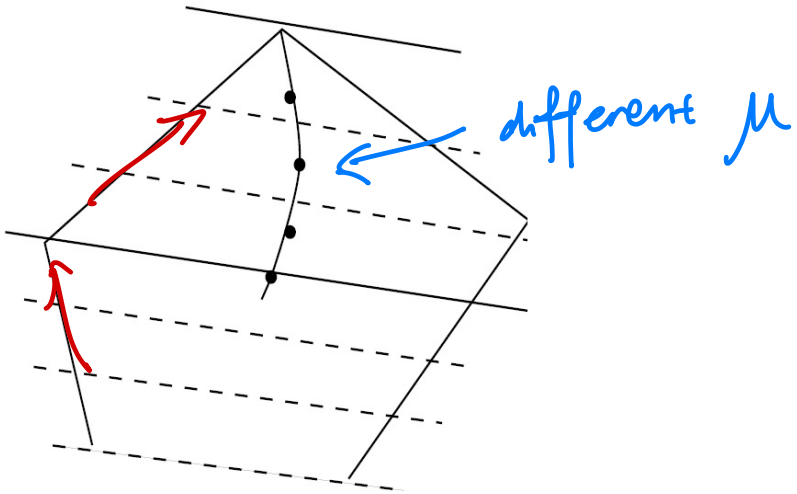


Figure 1: Central path in an interior point method

- In the simplex method, we only search among the extreme points (at the boundary of the polyhedron).
- In the interior point method, we start in the interior of the feasible region.
- Until we reach an optimal solution, we keep $\boldsymbol{x} > 0$ and $\boldsymbol{s} > 0$.

Nonlinear Programming

In general, we can write a nonlinear optimization problem as:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & f(\boldsymbol{x}) \\ \text{s.t.} \quad & \boldsymbol{x} \in \Omega \end{aligned}$$

We call Ω the feasible set and $\boldsymbol{x} \in \Omega$ are feasible points.

Recap: Global and Local Optima

Let $\Omega \subseteq \mathbb{R}^n$ be a nonempty set and let $f : \Omega \to \mathbb{R}$ be given.

We define $B_\varepsilon(\boldsymbol{y}) := \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x} - \boldsymbol{y}\| < \varepsilon\}$ to be the open ball in $\mathbb{R}^n$ with center $\boldsymbol{y}$ and radius $\varepsilon > 0$.

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The point $\boldsymbol{x}^* \in \mathbb{R}^n$ is said to be a:

- local minimizer, if $\boldsymbol{x}^* \in \Omega$ and there is $\varepsilon > 0$ with $f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*)$ for all $\boldsymbol{x} \in \Omega \cap B_\varepsilon(\boldsymbol{x}^*)$.



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- strict local minimizer, if $\boldsymbol{x}^* \in \Omega$ and there exists $\varepsilon > 0$ such that $f(\boldsymbol{x}) > f(\boldsymbol{x}^*)$ for all $\boldsymbol{x} \in (\Omega \cap B_\varepsilon(\boldsymbol{x}^*)) \backslash \{\boldsymbol{x}^*\}$.



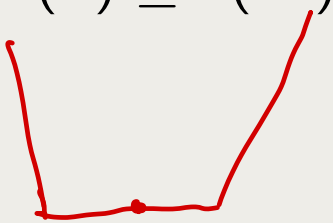
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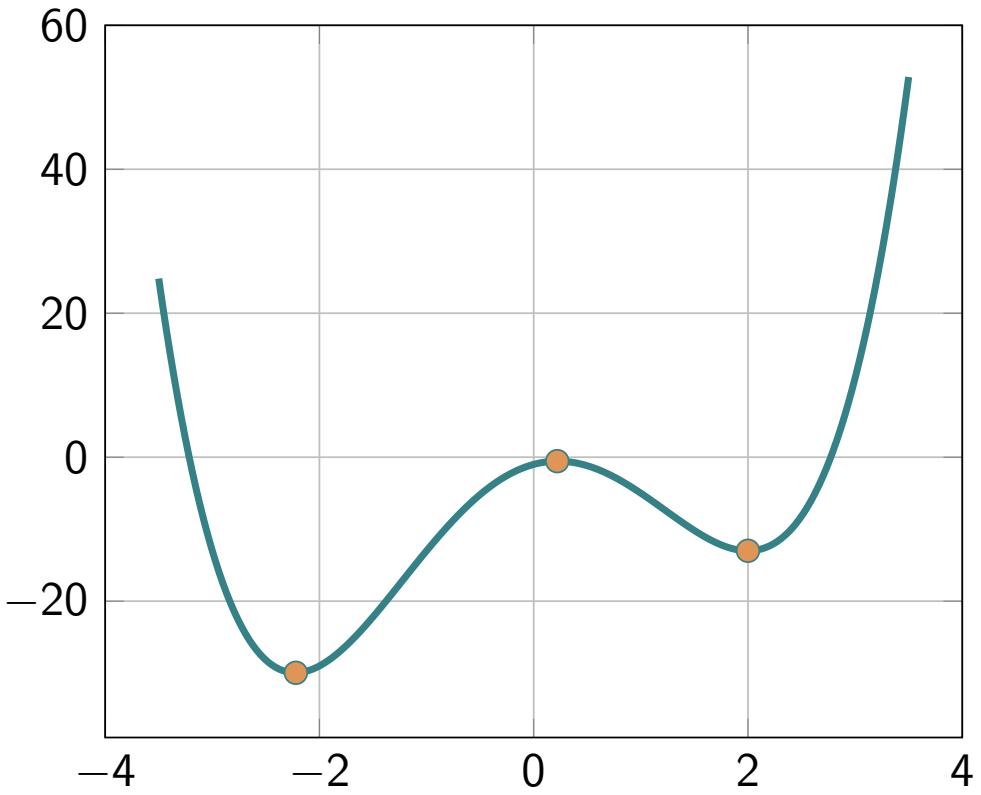
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Example: Minimizer

We consider the unconstrained problem

$$\text{minimize}_{x\in\mathbb{R}}\ f(x) := x^4 - 9x^2 + 4x - 1.$$



Review: Gradient and First-order Taylor Expansion

Assume $f(\boldsymbol{x}) = f(x_1, x_2, ..., x_n)$ is continuously differentiable. Then the gradient of f is denoted by (an $n \times 1$ vector):

$$\nabla f(\boldsymbol{x}) = \left(\frac{\partial f}{\partial x_1}; \frac{\partial f}{\partial x_2}; ...; \frac{\partial f}{\partial x_n} \right)$$

$$\begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

The first-order Taylor expansion yields:

$$f(\boldsymbol{x} + t\boldsymbol{d}) = f(\boldsymbol{x}) + t\nabla f(\boldsymbol{x})^\top \boldsymbol{d} + o(t), \quad t \to 0$$

linear approx.

$f(n) = o(g(n))$, $n \to 0$: $\lim_{n \to 0} \frac{f(n)}{g(n)} = 0$

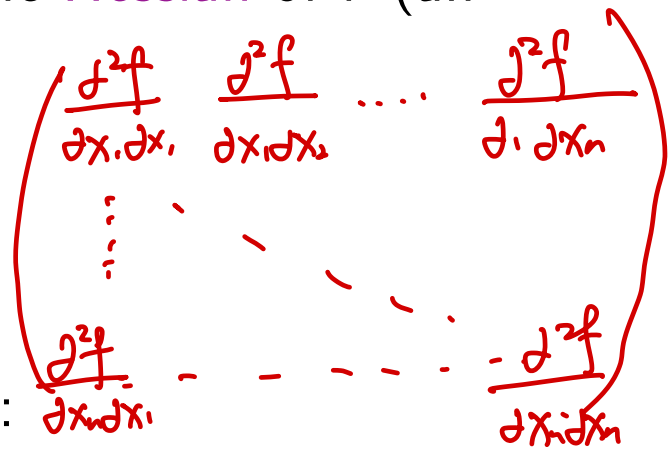
for any $c > 0$, $\exists N$ s.t. $f(n) < c\, g(n)$ for any $n < \frac{1}{N}$

Expansion

If f is twice continuously differentiable, then the Hessian of f (an $n \times n$ matrix) is given by:

symmetric

$$\nabla^2 f(\boldsymbol{x}) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)_{i,j}$$



By a second-order Taylor expansion, we obtain:

$$f(\boldsymbol{x} + t\boldsymbol{d}) = f(\boldsymbol{x}) + t\nabla f(\boldsymbol{x})^\top \boldsymbol{d} + \frac{1}{2} t^2 \boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x}) \boldsymbol{d} + o(t^2)$$

second-order approx.

Example

Suppose:

$$f(\underbrace{x_1, x_2, x_3}_{x}) := x_1^2 + x_1 x_2 + x_1 e^{x_3} + x_2 \log x_3$$

Then:

$$\nabla f(x) = \begin{pmatrix} 2x_1 + x_2 + e^{x_3} \\ x_1 + \log x_3 \\ x_1 e^{x_3} + \frac{x_2}{x_3} \end{pmatrix}$$

$$\nabla^2 f(x) = \begin{pmatrix} 2 & 1 & e^{x_3} \\ 1 & 0 & \frac{1}{x_3} \\ e^{x_3} & \frac{1}{x_3} & x_1 e^{x_3} - \frac{x_2}{x_3^2} \end{pmatrix}$$

Example

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Then:

$$\nabla f(\boldsymbol{x}) = \left(2x_1 + x_2 + e^{x_3}, x_1 + \log x_3, x_1 e^{x_3} + \frac{x_2}{x_3}\right)^\top$$

and

$$\nabla^2 f(\boldsymbol{x}) = \begin{pmatrix} 2 & 1 & e^{x_3} \\ 1 & 0 & \frac{1}{x_3} \\ e^{x_3} & \frac{1}{x_3} & x_1 e^{x_3} - \frac{x_2}{x_3^2} \end{pmatrix}$$

Optimality Conditions

Optimality Conditions

In the following, we first study what conditions an optimal solution has to satisfy:

- ⇝ First- and second-order optimality conditions.
- We will first start with local optimal solutions.

Optimality Conditions: Unconstrained Problems

Consider the unconstrained problem, $\Omega = \mathbb{R}^n$

$$\text{minimize}_{\boldsymbol{x}} f(\boldsymbol{x})$$

What is a necessary optimality condition for local minimizers?

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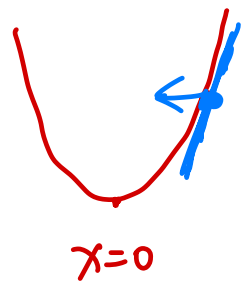
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What is a necessary optimality condition for local minimizers?

Claim: If f is differentiable and $\boldsymbol{x}$ is a local minimizer, then

$$\nabla f(\boldsymbol{x}) = \mathbf{0}.$$

$f(x) = x^2$ $\quad \nabla f(x) = 2x$



Intuition: if $\nabla f(x) \neq 0$, we can find a direction at x s.t the first-order approx. of f (denote it $\tilde{f}$) decreases. Since $\tilde{f}$ is close to f by taylor expansion. the going this direction also dreases f.

Optimality Conditions: Unconstrained Problems

Consider the unconstrained problem, $\Omega = \mathbb{R}^n$

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What is a necessary optimality condition for local minimizers?

Claim: If f is differentiable and $\boldsymbol{x}$ is a local minimizer, then

$$\nabla f(\boldsymbol{x}) = \mathbf{0}.$$

Proof: If $\nabla f(\boldsymbol{x}) \neq \mathbf{0}$, then we can find a vector $\boldsymbol{d}$ with $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$. Therefore, by Taylor expansion:

$$f(\boldsymbol{x} + t\boldsymbol{d}) = f(\boldsymbol{x}) + t\nabla f(\boldsymbol{x})^\top \boldsymbol{d} + o(t), \quad t \to 0.$$

By choosing t small enough, we can find a point $\bar{\boldsymbol{x}} = \boldsymbol{x} + t\boldsymbol{d}$ in the neighborhood of $\boldsymbol{x}$ such that $f(\bar{\boldsymbol{x}}) < f(\boldsymbol{x})$.

First-Order Optimality Conditions

First-Order Necessary Conditions (FONC)

First-Order Necessary Conditions

If $\boldsymbol{x}^*$ is a local minimizer of the unconstrained problem $\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$, then we must have $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$.

Remark:

- Points $\boldsymbol{x}$ with $\nabla f(\boldsymbol{x}) = \mathbf{0}$ are all candidates for local minima

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Example: $f(\boldsymbol{x}) = x_1^2 - x_1x_2 + x_2^2 - 3x_2$.

$$\nabla f(x) = \begin{pmatrix} 2x_1 - x_2 \\ -x_1 + 2x_2 - 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

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The FONC is:

$$2x_1 - x_2 = 0, \quad -x_1 + 2x_2 = 3.$$

There is a unique solution $(x_1 = 1,\ x_2 = 2)$, which turns out to be the global minimizer of f.

Example: Least Squares Problem

Assume a variable $\boldsymbol{y}$ is affected by n factors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^m$. We know that they approximately have a linear relationship:

$$\boldsymbol{y} \approx \beta_1 \boldsymbol{x}_1 + \cdots + \beta_n \boldsymbol{x}_n = \begin{bmatrix} | & & | \\ \boldsymbol{x}_1 & \cdots & \boldsymbol{x}_n \\ | & & | \end{bmatrix} \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}.$$

$\triangleq X$ $\triangleq \beta$

Given $\boldsymbol{X}$ and $\boldsymbol{y}$, how to find parameters β?

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Given $\boldsymbol{X}$ and $\boldsymbol{y}$, how to find parameters $\boldsymbol{\beta}$?

- Ideally, we want to find $\beta = (\beta_1, \ldots, \beta_n)^\top$ such that $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta}$. However, this may not be possible (the equation $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta}$ is an overdetermined linear system with $m > n$).
- Therefore, we want to find $\boldsymbol{\beta}$ such that $\boldsymbol{X}\boldsymbol{\beta}$ is close to $\boldsymbol{y}$.

Example: Least Squares — Continued

Try to minimize the sum of the squared errors:

$$\text{minimize}_{\beta} \quad \sum_{i=1}^{m} \left(y_i - \sum_{j=1}^{n} \beta_j x_{ij} \right)^2$$

The matrix form of this problem is:

quadratic function in β.

$$\text{minimize}_{\beta} \quad \|\boldsymbol{X}\beta - \boldsymbol{y}\|^2 = \beta^\top \boldsymbol{X}^\top \boldsymbol{X}\beta - 2\beta^\top \boldsymbol{X}^\top \boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{y}$$

vector

$$\|X\beta - y\|^2 = (X\beta - y)^\top (X\beta - y)$$

Example: Least Squares — Continued

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Facts:

- If $f(\boldsymbol{x}) = \boldsymbol{c}^\top \boldsymbol{x}$, then $\nabla f(\boldsymbol{x}) = \boldsymbol{c}$.
- If $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x}$ ($\boldsymbol{M}$ is symmetric), then: $\nabla f(\boldsymbol{x}) = 2\boldsymbol{M}\boldsymbol{x}$.

if $M=I$, $f(x) = \|x\|^2$

$f(x) = c^T x = \sum c_i x_i$

$\nabla f(x) = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = c$

$f(x) = x^T M x = \sum_i \sum_j x_i M_{ij} x_j$

$\frac{\partial f(x)}{\partial x_k} = \sum_{j=1}^{n} M_{kj} x_j + \sum_{i=1}^{n} x_i M_{ik}$.

if symmetric, $M_{kj} = M_{jk} \rightarrow \frac{\partial f(x)}{\partial x_k} = \sum_j 2 M_{kj} x_k$

Example: Least Squares — Continued

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$= (X^T y)^T \beta$

Facts:

$\nabla f(\beta) = 2X^T X\beta - 2X^T y$

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The FONC for the least squares problem is:

$$\boldsymbol{X}^\top \boldsymbol{X}\beta - \boldsymbol{X}^\top \boldsymbol{y} = \boldsymbol{0}.$$

Solutions of this equation are the candidates for local minimizer.

Example: Linear Regression via Least Squares

Suppose we have n given points (x_i, y_i) in $\mathbb{R}^2$.

Task: We want to find a linear mapping $y = \beta_1 x + \beta_2$ with coefficients $\boldsymbol{\beta} = (\beta_1, \beta_2)^\top$ that best fit the points (x_i, y_i):

$$y_i \approx \beta_1 x_i + \beta_2 \quad \forall\ i = 1, \ldots, n.$$

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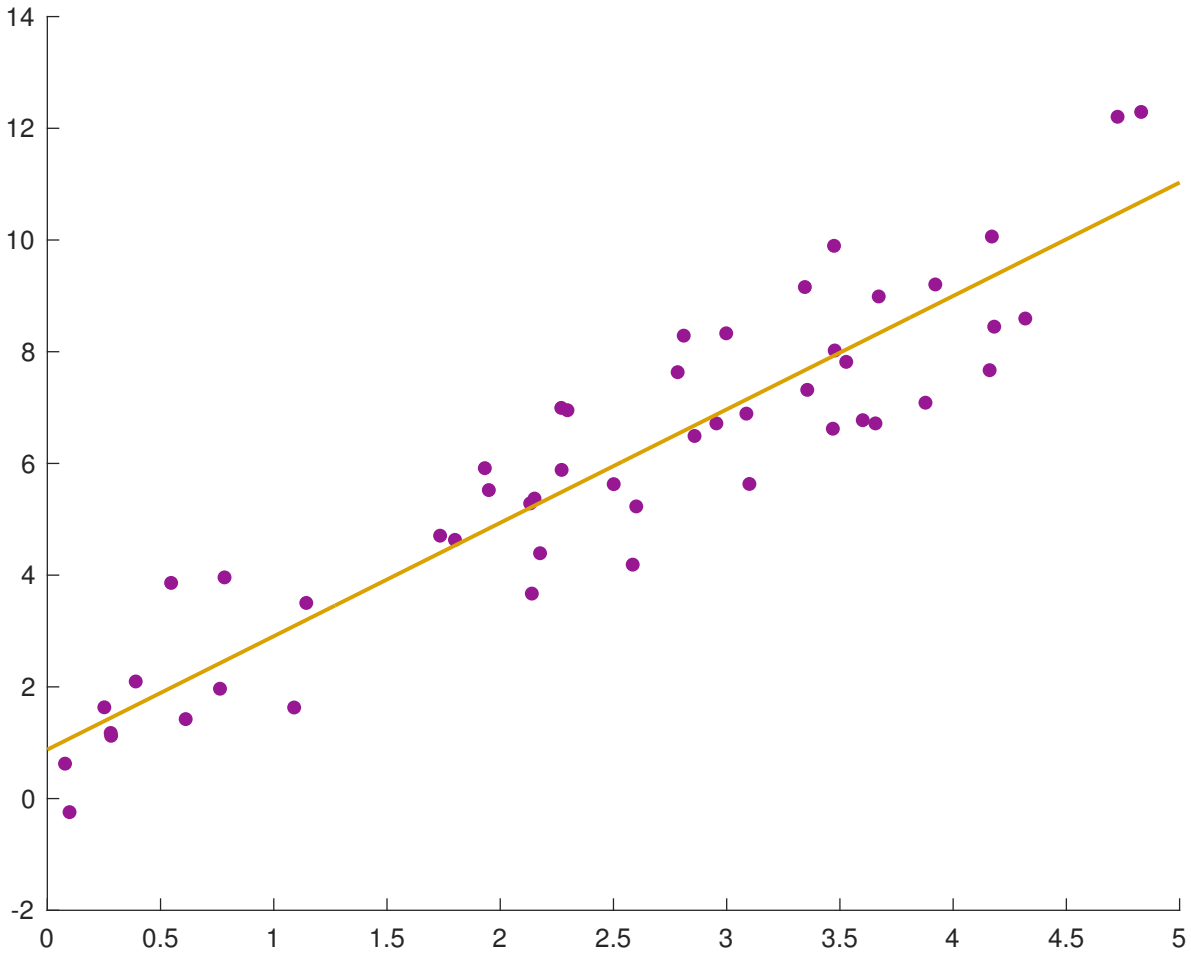
$$y_i \approx \beta_1 x_i + \beta_2 \quad \forall\ i = 1, \ldots, n.$$

- Setting $\boldsymbol{x} = (x_1, \ldots, x_n)^\top$ and $\boldsymbol{y} = (y_1, \ldots, y_n)^\top$, this can be written as

$$\boldsymbol{y} \approx \begin{bmatrix} | & 1 \\ \boldsymbol{x} & \vdots \\ | & 1 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}. \qquad \min_{\beta} \|y - x\beta\|^2$$

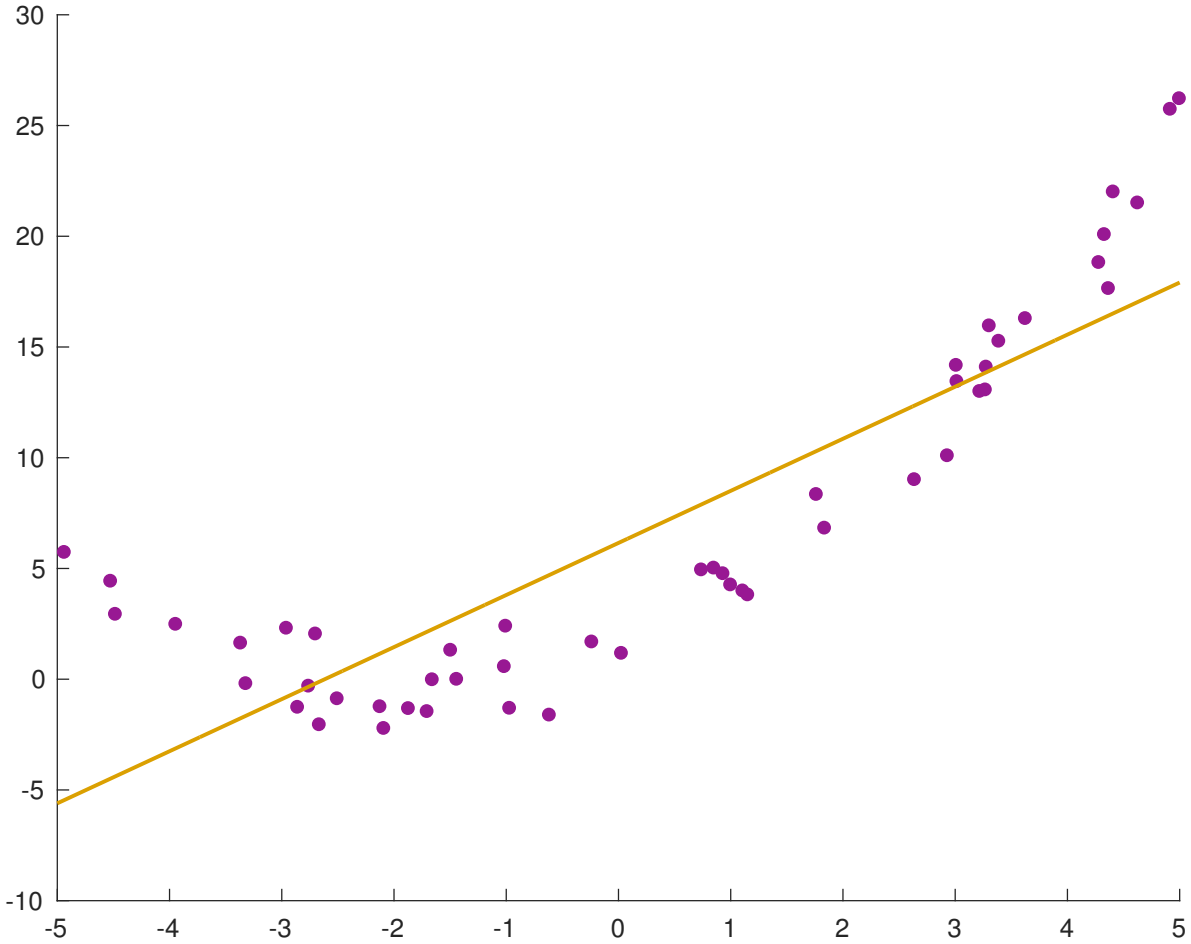
- Solve via least squares formulation.

Linear Regression: Visualization — I



- Linear regression based on 50 randomly selected points.

Linear Regression: Visualization — II



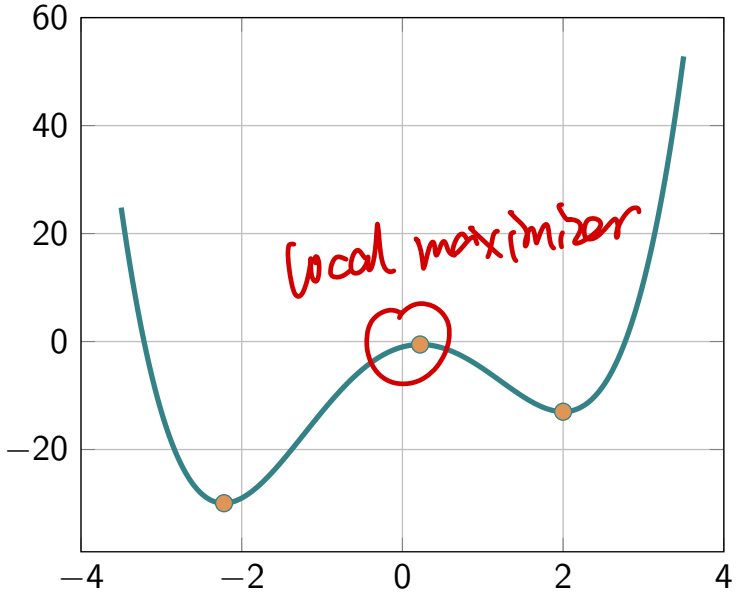
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FONC is Not Sufficient

Example: $f(x) := x^4 - 9x^2 + 4x - 1$. The FONC requries

$$f'(x) = 4x^3 - 18x + 4 = 0$$

with solutions $x_1 = -1 - \sqrt{6}/2$, $x_2 = -1 + \sqrt{6}/2$, and $x_3 = 2$.

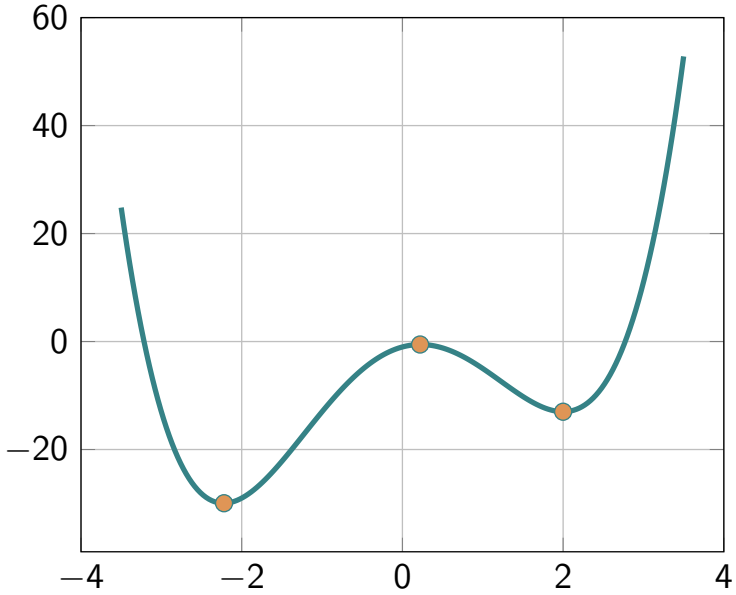


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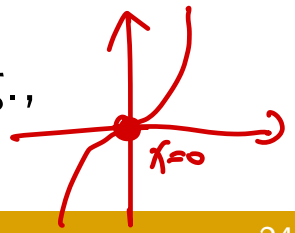
with solutions $x_1 = -1 - \sqrt{6}/2$, $x_2 = -1 + \sqrt{6}/2$, and $x_3 = 2$.



FONC is not sufficient

- Each local maximum also satisfies the FONC!
- It could be neither a local minimum nor maximum (e.g., $f(x) = x^3$).

prove it in the same way as FONC for local minimizer.



Second-Order Necessary Conditions

Second-Order Necessary Condition

Consider the Taylor expansion again but to the 2nd order (assuming f is twice continuously differentiable):

$$f(\boldsymbol{x}+t\boldsymbol{d}) = f(\boldsymbol{x}) + t\nabla f(\boldsymbol{x})^\top \boldsymbol{d} + \frac{1}{2}t^2 \boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x})\boldsymbol{d} + o(t^2).$$

$\nabla f(\boldsymbol{x}) = 0$

When the first-order necessary condition holds, we have:

$$f(\boldsymbol{x}+t\boldsymbol{d}) = f(\boldsymbol{x}) + \frac{1}{2}t^2 \boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x})\boldsymbol{d} + o(t^2).$$

In order for $\boldsymbol{x}$ to be a local minimizer, we also need $\boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x})\boldsymbol{d}$ to be nonnegative for every $\boldsymbol{d} \in \mathbb{R}^n$.

< 0 for t small enough

why: if $\exists\, d$ s.t. $d^\top \nabla^2 f(x) d < 0$, then when t small enough. the remaining term $o(t^2)$ is dominated by the negative $\frac{1}{2}t^2 d^\top \nabla f(x) d$.

Second-Order Necessary Condition (SONC)

Theorem: Second-Order Necessary Conditions

f is diff.

If $\boldsymbol{x}^*$ is a local minimizer of f, then it holds that:

1. $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$;
2. For all $\boldsymbol{d} \in \mathbb{R}^n$: $\boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x}^*) \boldsymbol{d} \geq 0$.

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Definition: Semidefinite Matrices

We call a (symmetric) matrix $\boldsymbol{A}$ positive (negative) semidefinite (PSD/NSD) if and only if for all $\boldsymbol{x}$ we have $\boldsymbol{x}^\top \boldsymbol{A} \boldsymbol{x} \geq 0$ (≤ 0).

Remark:

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Remark:

- Therefore, the second-order necessary condition requires the Hessian matrix at x^* to be PSD.
- In the one-dimensional case, this is equivalent to $f''(x^*) \geq 0$.

Here are some useful facts about PSD matrices:

- We usually only talk about PSD properties for symmetric matrices.
- If a matrix $\boldsymbol{A}$ is not symmetric, we use $\frac{1}{2}(\boldsymbol{A}+\boldsymbol{A}^\top)$ to define the PSD properties (because $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} = \frac{1}{2}\boldsymbol{x}^\top(\boldsymbol{A}+\boldsymbol{A}^\top)\boldsymbol{x}$).
- A symmetric matrix is PSD if and only if all the eigenvalues are nonnegative.
- For any matrix $\boldsymbol{A}$, $\boldsymbol{A}^\top\boldsymbol{A}$ is a (symmetric) PSD matrix.

square

symmetric

- $(A^TA)^T = A^TA \rightarrow A^TA$ is symmetric
- $\forall x \quad x^TA^TAx = \|Ax\|^2 \geq 0$

$\rightarrow A^TA$ is PSD

Example Continued

For $f(x) := x^4 - 9x^2 + 4x - 1$, the second-order condition is:

$$f''(x) = 12x^2 - 18 \geq 0.$$

Only $x_1 = -1 - \sqrt{6}/2$ and $x_3 = 2$ satisfy the condition. But for the point $x_2 = -1 + \sqrt{6}/2$, we obtain $f''(x_2) = 12(1 - \sqrt{6}) < 0$ (thus, x_2 is not a local minimizer).

In the example of least squares problem, we use the following fact:

- If $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x}$ ($\boldsymbol{M}$ is symmetric), then $\nabla^2 f(\boldsymbol{x}) = 2\boldsymbol{M}$.

Therefore, the Hessian matrix in that problem is $2\boldsymbol{X}^\top \boldsymbol{X}$, which is always a PSD matrix. Therefore, the SONC always holds.

SONC is Not Sufficient

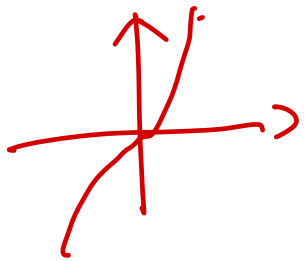
However, even if both the first- and second-order necessary conditions hold, we still can not guarantee that the candidate is a local minimum.

$f'(x) = 3x^2$

$f''(x) = 6x$

Example: Consider $f(x) = x^3$ at 0.

- $f'(0) = f''(0) = 0$, thus FONC and SONC hold.
- But 0 is not a local minimum



$f(x) < f(0)$

$\forall x < 0$

SONC is Not Sufficient

However, even if both the first- and second-order necessary conditions hold, we still can not guarantee that the candidate is a local minimum.

Example: Consider $f(x) = x^3$ at 0.

- $f'(0) = f''(0) = 0$, thus FONC and SONC hold.
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- The SONC can used to verify that a point $\boldsymbol{x}$ with $\nabla f(\boldsymbol{x}) = \mathbf{0}$ is **not** a local minimizer.

Second-Order Sufficient Conditions

Second-Order Sufficient Condition (SOSC)

Theorem: Second-Order Sufficient Conditions

Let f be twice continuously differentiable. If x^* satisfies:

1. $\nabla f(x^*) = 0$;
2. For all $d \in \mathbb{R}^n \backslash \{0\}$: $d^\top \nabla^2 f(x^*) d > 0$;

then x^* is a strict local minimum of f.

Example.



$f(x) = x^2$

$f'(x) = 2x$

$f''(x) = 2 > 0$

$\Rightarrow x = 0$ is a strictly local minimizer

Definition: Definite Matrix

We call a (symmetric) matrix $\boldsymbol{A}$ positive (negative) definite (PD/ND) if and only if for all $\boldsymbol{x} \neq \boldsymbol{0}$ we have $\boldsymbol{x}^\top \boldsymbol{A} \boldsymbol{x} > 0$ (< 0).

Remark:

- Therefore, the second-order sufficient condition requires the Hessian matrix at x^* to be PD.
- In the one-dimensional case, this is equivalent to $f''(x^*) > 0$.

Lemma: Bounds and Eigenvalues

Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix. Then

$$\lambda_{\min}(A)\|x\|^2 \le x^\top A x \le \lambda_{\max}(A)\|x\|^2 \quad \forall\, x \in \mathbb{R}^n,$$

where $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ are the smallest and largest EV of A.

When $\nabla^2 f(x^*)$ is positive def., we have $\mu := \lambda_{\min}(\nabla^2 f(x^*)) > 0$ and $d^T \nabla^2 f(x^*) d \ge \mu \|d\|^2$ for all d.

Thus, for any x^* that satisfies the second-order sufficient condition, for any sufficiently small d:

$$f(x^* + d) = f(x^*) + \frac{1}{2} d^\top \nabla^2 f(x^*) d + o(\|d\|^2) > f(x^*)$$

which implies that x^* is a strict local minimizer.

For Maximization Problems

Our conditions are derived for minimization problems. For maximization problems, we just change the inequalities. Let $f \in C^2$.

Theorem: FONC for Maximization

If x^* is a local (unconstrained) maximizer of f, then we must have $\nabla f(x^*) = 0$.

Theorem: SONC for Maximization

If x^* is a local maximizer of f, then we must have (1) $\nabla f(x^*) = 0$; (2) $\nabla^2 f(x^*)$ is negative semidefinite.

Theorem: SOSC for Maximization

If x^* satisfies (1) $\nabla f(x^*) = 0$; (2) $\nabla^2 f(x^*)$ is negative definite, then x^* is a strict local maximizer.

Today's lecture

- FONC
- SONC
- SOSC

Next lecture

- More about Optimality Condition

Questions?